

EE236: Electronic Devices Lab Lab 0

Abhineet Agarwal, 22b1219

August 4, 2024

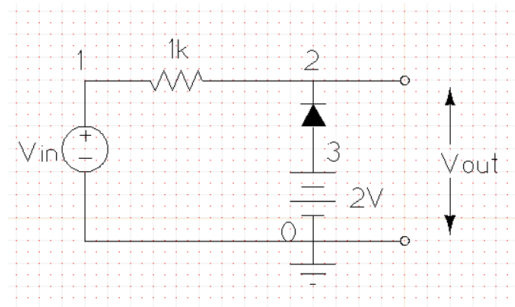
1 Shunt Clipper

1.1 Aim of the experiment

Simulating a shunt clipper circuit in different configurations and observing the output waveform.

1.2

1.2.1 Design



1.2.2 Simulation Results

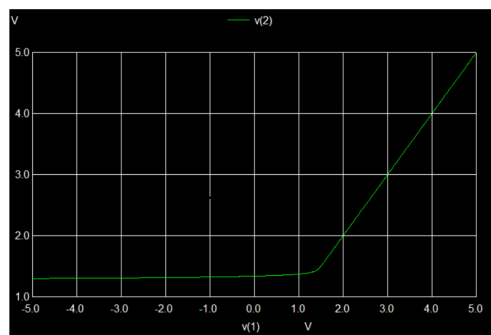


Figure 1: V_{out} vs V_{in}

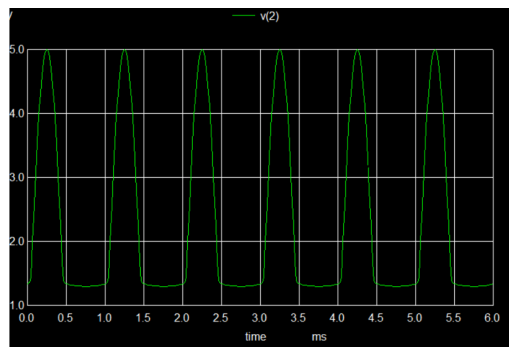
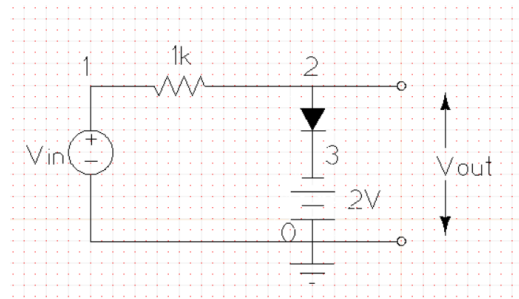


Figure 2: Output Waveform



1.3

1.3.1 Design

1.3.2 Simulation Results

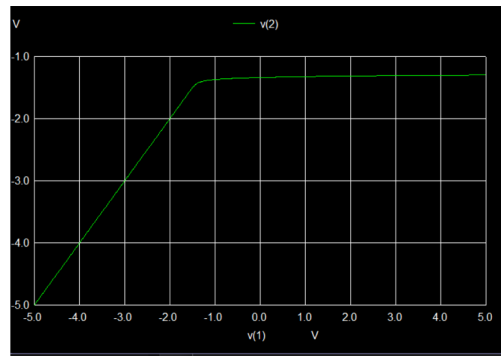


Figure 3: V_{out} vs V_{in}

1.4 Conclusion and Inference

This experiment demonstrated how to make shunt clippers with different input-output characteristics depending on the requirement.

1.5 Experiment completion status

Completed.

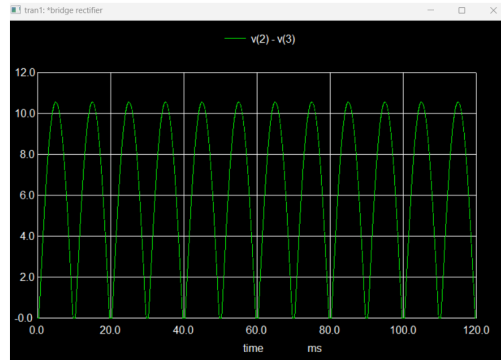


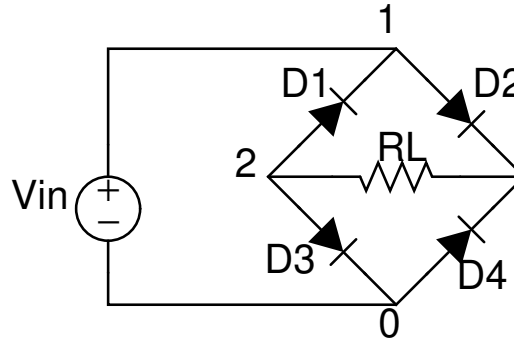
Figure 4: Output Waveform

2 Diode-based Bridge Rectifier

2.1 Aim of the experiment

Simulating a diode-based bridge rectifier circuit with a resistive load.

2.2 Design



2.3 Simulation Results

2.4 Conclusion and Inference

This experiment demonstrated how to construct a full-wave bridge rectifier using diodes.

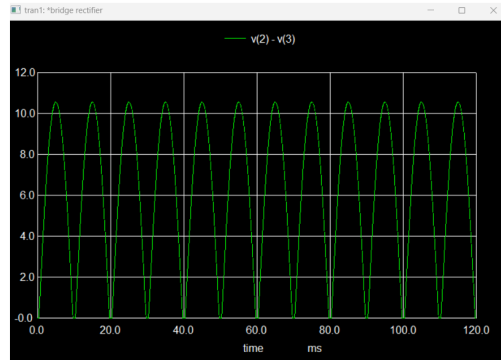


Figure 5: Output Waveform

2.5 Experiment completion status

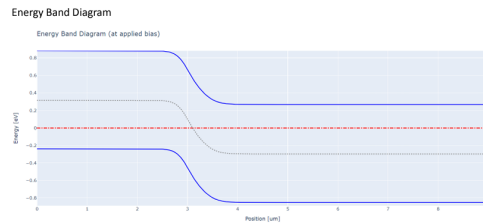
Completed.

3 PN Junction Simulation on Nanohub

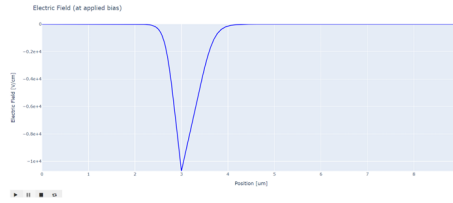
3.1 Aim of the experiment

Simulate a PN Junction on Nanohub and study how different quantities change within the device and with bias voltage.

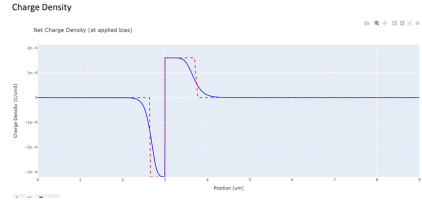
3.2 Energy Band Diagram



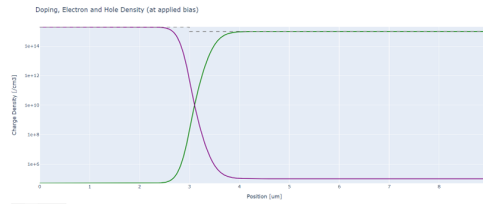
3.3 Electric Field



3.4 Charge Density



3.5 Total Density



3.6 Effect on changing the Bias Voltage

On increasing the bias voltage, the depletion layer width decreases.

- The slope of the energy band diagram decreases.
- The electric field in the depletion region decreases.
- The charge density in the depletion region decreases.
- The electron and hole density increases across the device.

3.7 Experiment completion status

Completed.